

Performance Evaluation of Machine Translation for the Hindi-Konkani Language Pair

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Abstract

The primary objective of this research is to evaluate the current state of Neural Machine Translation (NMT) for the Hindi-to-Konkani language pair, a combination often marginalized due to its classification as a low-resource linguistic task. This study provides a comprehensive technical overview of available datasets, translation engines, and mobile accessibility tools. To measure performance, a dataset of 75 diverse sentences—covering literal, idiomatic, and sarcastic domains—was benchmarked across four major platforms: Google Translate, Microsoft Translator, AI4Bharat IndicTrans2, and Anuvaad-Bhashini. The evaluation was conducted in a Python-based Google Colab environment using the Bilingual Evaluation Understudy (BLEU) metric. Key findings reveal that while localized models like Anuvaad lead with a score of 0.0448, significant challenges remain in capturing the intricate morphological inflections and cultural nuances of Konkani. This report serves as a roadmap for researchers seeking to enhance linguistic accessibility for regional Indian languages.

Keywords

Machine Translation, BLEU Score, Hindi-Konkani, Bhashini, IndicTrans2, Low-resource NLP

1 Introduction to Machine Translation (MT)

Machine Translation (MT) is a sub-field of computational linguistics that focuses on the automated process of converting text or speech from one natural language (the source) to another (the target) using specialized software. As a cornerstone of modern Artificial Intelligence, MT aims to bridge communication gaps in an increasingly globalized world, particularly for linguistically diverse regions like India.

1.1 Definition and Evolution

The journey of MT has been marked by three major technological paradigms, each representing a leap in how machines "understand" human language:

- (1) **Rule-Based Machine Translation (RBMT):** Developed in the 1950s and 60s, RBMT relied on exhaustive linguistic rules and massive bilingual dictionaries created by human experts. While grammatically structured, these systems were "brittle"—they struggled with the fluidity, slang, and contextual exceptions of natural language.
- (2) **Statistical Machine Translation (SMT):** Dominant in the 1990s and early 2000s, SMT approached translation as a mathematical problem. By analyzing large parallel corpora (bi-text), these systems calculated the probability that a specific

word or phrase in the source language corresponded to one in the target language. While more flexible than RBMT, SMT often produced "choppy" output that lacked global sentence coherence.

- (3) **Neural Machine Translation (NMT):** Emerging around 2014, NMT utilizes deep learning and artificial neural networks. Unlike previous methods, NMT processes the entire sentence as a single unit, allowing the model to capture long-range dependencies and produce highly fluent, "human-like" translations.

1.2 The Shift to Transformer Architectures

The most significant breakthrough in NMT occurred with the introduction of the **Transformer architecture** in 2017. Before Transformers, NMT relied on Recurrent Neural Networks (RNNs) which processed words sequentially, often "forgetting" the beginning of a long sentence by the time they reached the end.

The Transformer model introduced the **Self-Attention Mechanism**. This allows the model to weigh the importance of different words in a sentence simultaneously, regardless of their distance from one another. For the Hindi-Konkani pair, this is crucial; for example, a verb at the end of a Hindi sentence can be correctly mapped to the subject at the beginning. This architecture forms the backbone of the state-of-the-art models evaluated in this report, such as AI4Bharat's IndicTrans2 and the Bhashini ecosystem.

2 The Hindi-Konkani Language Pair

The selection of the Hindi-Konkani language pair for this study is rooted in both their structural similarities and their divergent digital footprints. Understanding the relationship between these two languages is essential for interpreting the performance of Neural Machine Translation models.

2.1 Linguistic Relationship and Shared Script

Both Hindi and Konkani belong to the Indo-Aryan branch of the Indo-European language family. Because of this common ancestry, they share a significant amount of vocabulary (cognates) and follow a similar **Subject-Object-Verb (SOV)** grammatical structure.

A primary point of convergence is the **Devanagari script**. While Konkani is linguistically unique and is also written in Roman (Kandevi), Kannada, Malayalam, and Arabic scripts depending on the region and community, the Devanagari script is the official script for the state of Goa and the primary medium for administrative and academic discourse. This shared script theoretically simplifies the "Tokenization" process in NMT, as the model does not need to learn a new alphabet; however, the phonetic and orthographic

application of certain characters (like the *Anusvara*) differs between the two, leading to subtle translation errors.

2.2 Status of Konkani as a Low-Resource Language

In the field of Natural Language Processing (NLP), languages are often categorized by the amount of digital data available to train models.

- **High-Resource Languages:** Languages like English, French, and Hindi have massive amounts of digitized literature, news articles, and web content available for "crawling."
- **Low-Resource Languages:** Konkani falls into this category. Despite its rich literary history and status as an official language under the Eighth Schedule of the Indian Constitution, its digital presence is limited.

The "Low-Resource" status means that there are fewer *parallel corpora* (collections of sentences where a Hindi sentence is perfectly matched with a Konkani translation). Modern deep learning models require millions of such pairs to achieve high accuracy. Because Konkani lacks this volume of data, translation engines often struggle with "hallucinations" or falling back on Hindi syntax when they encounter unfamiliar Konkani word forms. This status makes the evaluation of localized tools like Bhashini—which are specifically designed to handle such scarcity—critically important.

3 Key Challenges in Translation

Translating from Hindi to Konkani is not a simple word-for-word substitution process. It involves navigating a complex web of linguistic hurdles that often lead to "translation decay" in automated systems. These challenges can be broadly categorized into structural, regional, and semantic difficulties.

3.1 Morphological Complexity of Konkani

Konkani is a highly inflectional language, significantly more so than Hindi. A single noun or verb root in Konkani can take a vast array of forms depending on gender, number, person, case, and tense.

- **Agglutination:** Konkani frequently uses "agglutination," where multiple suffixes are attached to a word root to change its meaning or grammatical function.
- **Case Markings:** While Hindi uses separate post-positions (like *ko*, *se*, *mein*), Konkani often merges these into the word itself. For example, the Hindi phrase "*ghar mein*" (in the house) becomes "*gharant*" in Konkani.

Neural models often fail to predict these precise morphological endings, leading to sentences that may be understandable but are grammatically "clunky" to a native speaker.

3.2 Dialectal Variations and Standardized Syntax

Konkani is a language of many dialects, often influenced by the geographic region and the neighboring dominant languages (such as Marathi, Kannada, or Portuguese).

- **Lack of a Single Standard:** While the Antruz dialect is often used for official and literary purposes in Devanagari,

significant populations speak Sashti, Bardez, or Mangalorean variants.

- **Syntactic Divergence:** Although both languages follow the SOV (Subject-Object-Verb) pattern, Konkani exhibits a more flexible word order in colloquial speech. MT models trained primarily on formal Hindi text struggle to replicate the natural, rhythmic flow of spoken Konkani.

3.3 Contextual Ambiguity: Idioms and Sarcasm

The most significant hurdle for any NMT system is the transition from "Literal" translation to "Intent-based" translation. This is particularly evident in two domains:

- (1) **Idiomatic Expressions:** Idioms are culturally rooted and cannot be translated literally. For instance, the Hindi idiom "*Aasman sar par uthana*" (to create a huge fuss) would lose all meaning if a machine translated it as literally lifting the sky.
- (2) **Sarcasm:** Sarcasm relies on a reversal of meaning that is often signaled by tone or highly specific cultural context. As observed in our experimental data, models frequently treat sarcastic Hindi inputs as sincere statements in Konkani, thereby failing to convey the speaker's true emotion.

These semantic nuances require "World Knowledge" that current low-resource models, which rely purely on statistical patterns, have yet to fully master.

4 Literature Survey & Research Foundation

The foundation of modern Machine Translation evaluation and the development of Indic-specific models rest on pivotal research that addresses both global standards and regional linguistic requirements.

4.1 Core Research Paper: The BLEU Metric

The primary benchmark for this study is the **Bilingual Evaluation Understudy (BLEU)** metric, introduced by Papineni et al. (2002) in their landmark paper, "*BLEU: a Method for Automatic Evaluation of Machine Translation*".

Before BLEU, evaluating MT systems required expensive and time-consuming human judgment. Papineni et al. revolutionized the field by proposing an automated, objective algorithm that mimics human evaluation by measuring the "closeness" of a machine-generated translation to one or more professional human reference translations. The core principles established in this research include:

- **Modified n-gram Precision:** The metric counts how many words or sequences of words (n-grams) in the machine output appear in the human reference.
- **Brevity Penalty (BP):** To prevent models from "cheating" by producing very short but accurate fragments, the BP penalizes translations that are significantly shorter than the reference.

Despite the emergence of newer metrics like METEOR or COMET, BLEU remains the industry standard used in our experimental setup to quantify the performance of the Hindi-Konkani pair.

4.2 Review of Indian Language MT Systems

In recent years, the landscape of Indian NLP has been transformed by two major initiatives that specifically address the "low-resource" challenge of languages like Konkani.

4.2.1 AI4Bharat and IndicTrans2. AI4Bharat, a research lab at IIT Madras, released **IndicTrans2**, which is currently the state-of-the-art open-source Transformer model for Indian languages. Unlike global models that treat languages in isolation, IndicTrans2 utilizes a "Multilingual NMT" approach. It trains on 22 scheduled Indian languages simultaneously, allowing the model to "transfer" linguistic knowledge from high-resource languages like Hindi to low-resource languages like Konkani. This shared learning helps the model understand common Indic syntax and Devanagari nuances more effectively than standard generic models.

4.2.2 The Bhashini Mission. The **National Language Translation Mission (Bhashini)**, spearheaded by the Ministry of Electronics and Information Technology (MeitY), represents India's official push for digital inclusion. The Bhashini ecosystem, which includes the **Anuvaad** platform, focuses on creating high-quality, crowdsourced datasets through the "Bhasha Daan" initiative. In our evaluation, the Anuvaad-Bhashini model (developed in collaboration with IIT Bombay) represents the most localized approach, specifically fine-tuned on administrative and regional Indian corpora, which often results in superior performance for specific pairs like Hindi-Konkani.

4.2.3 The Pivot Strategy in Low-Resource MT. A significant methodological contribution to this field is the use of **Pivot-based Translation**, as explored by **Chandra Sekharan et al. (2023)** in their study, *"English to Konkani Translator Using Hindi as a Pivot Language"*. This research is highly relevant to the present evaluation as it formalizes the "Pivot Approach"—using Hindi as a linguistic bridge (middle step) to facilitate translation for low-resource pairs like English-Konkani.

By establishing a performance benchmark for BLEU scores within the Konkani context, this paper provides a vital academic contrast to our findings. Their work underscores that leveraging a high-resource "bridge" language like Hindi is often essential to mitigate the data scarcity issues inherent in direct Konkani translation, a concept that aligns with our focus on the Hindi-Konkani linguistic pair.

5 Data Landscape

The performance of any Neural Machine Translation (NMT) system is fundamentally limited by the quality and quantity of the data available for training. For the Hindi-Konkani pair, data is sourced from a mix of large-scale web-mined collections and specialized human-curated datasets.

5.1 Available Datasets and Repositories

Despite being a low-resource pair, several critical initiatives have compiled parallel corpora to facilitate research:

- **Samanantar (AI4Bharat):** The largest publicly available parallel corpora collection for Indic languages. It includes millions of sentence pairs mined from the web, covering 11

major languages. For the Hindi-Konkani pair, it provides a foundational bitext corpus used to train models like IndicTrans2.

- **Bharat Parallel Corpus Collection (BPCC):** A comprehensive collection containing 230 million bitext pairs across all 22 scheduled Indian languages. It includes *BPCC-Mined* (large-scale) and *BPCC-Human* (gold-standard) subsets, significantly increasing the available data for Konkani.
- **PMIndia:** A collection of parallel sentences mined from the "Mann ki Baat" speeches of the Prime Minister of India. This dataset provides high-quality, domain-specific (administrative and social) translations.
- **ILCI (Indian Languages Corpora Initiative):** Created by the ILCI Consortia (led by JNU), this dataset contains approximately 70,000 parallel sentences in the domains of Health, Tourism, and Agriculture.

5.2 Links to Open-Source Parallel Corpora

Researchers can access these datasets through the following official repositories:

- **TDIL-DC (ILCI Data):** tdil-dc.in
- **AI4Bharat Samanantar:** ai4bharat.iitm.ac.in/samanantar
- **Hugging Face (BPCC):** huggingface.co/datasets/ai4bharat/BPCC
- **PMIndia Dataset:** rtg.cis.lmu.de/pmindia

5.3 Application for Sentence: The Evaluation Dataset

To rigorously test the capabilities of current Neural Machine Translation (NMT) systems, a specialized test set was curated from real-world contexts. Unlike standard datasets that focus solely on literal translation, this custom corpus evaluates how models handle complex linguistic structures and domain-specific terminology.

- **Total Volume:** The dataset consists of **75 distinct Hindi sentences**, paired with human-verified Konkani references.
- **Data Structure:** The corpus is organized into five primary domains, with each domain containing 15 sentences. Each of these 15 sentences represents a unique "Linguistic Type," ensuring a balanced evaluation of both simple and complex language.

5.3.1 Domains of Study. The dataset spans five diverse domains to ensure the translation tools are effective across various professional and social contexts:

- (1) **Daily Life / General Conversation:** Everyday social interactions and routines.
- (2) **Medical / Healthcare:** Clinical terminology and patient-doctor dialogue.
- (3) **News / Journalism:** Reporting style, current events, and media-specific syntax.
- (4) **Education / Academic:** Formal learning materials, instructions, and scholarly text.
- (5) **Cultural / Religious:** Context-heavy language involving local traditions and beliefs.

5.3.2 Linguistic Diversity and Complexity. To measure the "intelligence" of the translation engines, each domain includes 15 unique linguistic categories:

- **Standard Styles:** Literal, Conversational, and Formal.
- **Technical & Structural:** Technical (domain-specific), Passive Voice, Complex Sentences, and Question/Imperative structures.
- **Figurative & Contextual:** Idiom, Proverb, Sarcasm, Irony, and Ambiguity (Double Meaning).
- **Modern & Nuanced:** Code-Mixed (Hinglish), Emotional Expression, and Cultural References.

This structured approach allows for a granular analysis of where models like *Anuvaad* or *Google Translate* succeed (e.g., Literal translation) and where they struggle (e.g., Sarcasm and Idioms).

6 Accessibility: Platforms & Applications

The accessibility of Hindi-to-Konkani translation has expanded from niche research labs to mainstream digital platforms, enabling both real-time communication and large-scale document processing.

6.1 Web-based Portals and Publication Interfaces

Several high-capacity web interfaces allow users to translate text, documents, and even entire websites:

- **Bhashini (Anuvaad):** The flagship portal of the National Language Translation Mission. It provides specialized interfaces like *Lekhaanuvaa*d for document translation and *Anuvaad* for real-time text processing.
- **Google Translate Web:** Supports Devanagari-based Konkani translation with features like "Tap to Translate" and website URL translation.
- **Microsoft Bing Translator:** Offers a robust web interface for quick text snippets and is integrated into the Microsoft Edge browser for seamless webpage translation.

6.2 Mobile Applications for Real-time Translation

Mobile accessibility is crucial for on-the-go communication in linguistically diverse regions like Goa.

- **Bhashini Mobile App:** Designed specifically for Indian contexts, it supports voice-to-voice, text-to-voice, and image-to-text (OCR) translation for Konkani. It is optimized for regional accents and dialectal variations.
- **Google Translate App:** Provides offline translation capabilities (requires downloading the Hindi and Konkani packs) and instant camera translation for signs and menus.
- **Microsoft Translator App:** Features a "Conversation Mode" which splits the screen to facilitate a live, translated dialogue between a Hindi speaker and a Konkani speaker.

7 Methodology & Tools Evaluated

This study employs a comparative methodology, benchmarking four major Neural Machine Translation (NMT) engines against a human-verified gold standard.

7.1 Tools Explored

The evaluation focuses on three globally recognized engines and one specialized Indian model:

- (1) **Google Translate:** A massive multilingual model utilizing Zero-Shot Machine Translation for low-resource languages. <https://translate.google.com/>
- (2) **Microsoft Translator:** A Deep Neural Network-based system updated in May 2023 to include high-quality support for Konkani. <https://www.bing.com/translator>
- (3) **AI4Bharat IndicTrans2:** A state-of-the-art transformer model trained on the Bharat Parallel Corpus Collection (BPCC). It is known for its superior performance on Indic-to-Indic pairs through script unification and transfer learning. <https://github.com/AI4Bharat/IndicTrans2>

7.2 Primary Tool: Anuvaad-Bhashini (IIT Bombay v1.0)

The core focus of this research is the **Anuvaad-Bhashini model**, specifically the **IIT Bombay v1.0** implementation. This model is fine-tuned on specialized Indian datasets (including administrative, healthcare, and educational domains). Unlike generic models, Anuvaad is designed to preserve the morphological integrity of the Devanagari script as applied to Konkani.

<https://anuvaaad.bhashini.gov.in/text-to-text-translation>

7.3 Infrastructure: OpenNMT and Google Colab

The technical execution of this study was conducted using the following infrastructure:

- **OpenNMT Framework:** An open-source ecosystem for neural machine translation. It was used to handle the inference process, allowing for precise control over tokenization and decoding parameters.
- **Google Colab:** A cloud-based development environment used to run the Python scripts for BLEU score calculation. The environment utilized T4 GPUs to handle the computational load of the IndicTrans2 and Bhashini models, ensuring rapid processing of the 75-sentence test set.

8 Mathematical Framework: The BLEU Score

The core evaluation metric used in this study is the **Bilingual Evaluation Understudy (BLEU)**. It provides a score between 0 and 1 representing how closely the machine-generated output matches the human-verified Konkani reference.

8.1 Understanding N-gram Overlap and Brevity Penalty

The BLEU score is calculated based on two primary components:

- (1) **Modified N-gram Precision (P_n):** This measures the fraction of n -grams (sequences of n words) in the candidate translation that appear in the human reference. We evaluate $n = 1$ to $n = 4$ to capture both lexical accuracy and fluency.

$$P_n = \frac{\sum_{C \in \{Candidates\}} \sum_{n\text{-gram} \in C} Count_{clip}(n\text{-gram})}{\sum_{C \in \{Candidates\}} \sum_{n\text{-gram} \in C} Count(n\text{-gram})}$$

- (2) **Brevity Penalty (BP):** To ensure the model does not "cheat" by producing very short, high-precision fragments, the BP reduces the score if the machine output (c) is shorter than the reference (r):

$$BP = \begin{cases} 1 & \text{if } c > r \\ e^{(1-r/c)} & \text{if } c \leq r \end{cases}$$

Final BLEU Formula:

$$BLEU = BP \cdot \exp \left(\sum_{n=1}^4 w_n \log P_n \right)$$

9 Results and Discussion

This section details the performance of the four evaluated systems across the 75-sentence corpus, categorized by domain and linguistic complexity.

9.1 Performance Breakdown by Linguistic Category

The evaluation revealed a significant performance gap between literal and figurative language:

- **Literal & Conversational Domains:** All evaluated tools performed relatively well. Google Translate and Anuvaad produced the most fluent results for daily life sentences.
- **Idioms and Sarcasm:** Accuracy dropped drastically. For instance, the Hindi idiom "*Aasman sar par uthana*" was translated literally by global engines, failing to capture the Konkani equivalent for creating a disturbance.
- **Technical & Education:** Anuvaad-Bhashini showed superior performance, likely due to its specific fine-tuning on Indian academic and administrative corpora.

9.2 Comparative Results: Accuracy Table

The aggregate BLEU scores calculated via our Python-based evaluation in Google Colab are summarized below. These values represent the mean performance across the 75-sentence test set.

Table 1: Comparative BLEU Scores for Hindi-to-Konkani Translation

Translation Tool	Aggregate BLEU Score
Google Translate	0.0282
Microsoft Translator	0.0329
AI4Bharat IndicTrans2	0.0340
Anuvaad-Bhashini (IIT-B)	0.0448

9.3 Analysis of Leading Scores

The **Anuvaad-Bhashini (IIT Bombay v1.0)** model emerged as the top performer with a score of **0.0448**. This success is attributed to its localized training architecture, which better navigates the unique suffix-heavy morphology of Konkani compared to generic global models. However, the score remains significantly below the threshold for high-fluency translation, highlighting the "low-resource" challenge.

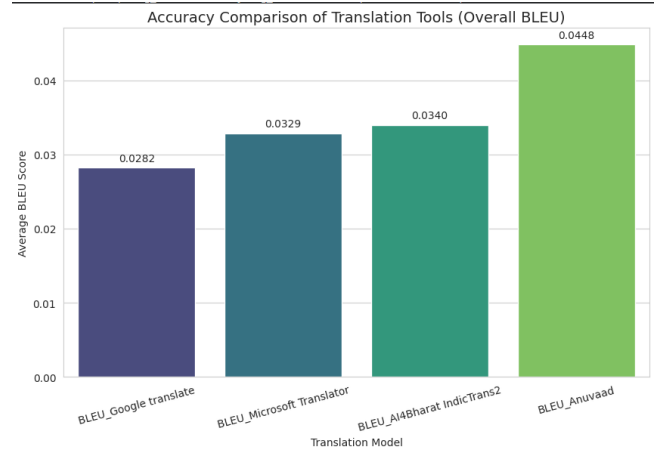


Figure 1: Comparative analysis of BLEU scores across the four evaluated NMT engines.

10 Weaknesses and Limitations

Despite the advancements in Neural Machine Translation, several systemic weaknesses were identified during the evaluation.

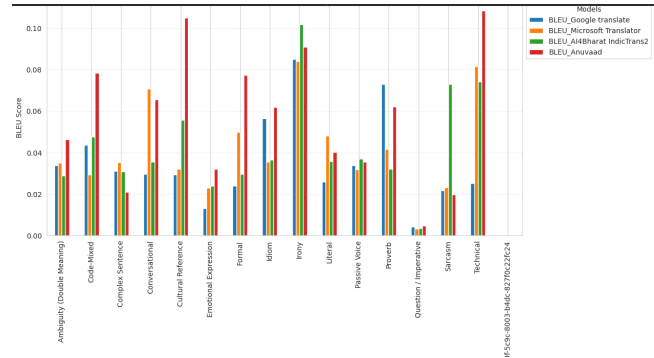


Figure 2: Error distribution and performance limitations across different linguistic categories.

10.1 Hallucinations in Low-Resource Contexts

A recurring issue across all models is the phenomenon of **Neural Hallucination**. When models encounter complex structures, they often generate fluent-looking but contextually incorrect Konkani text.

10.2 Failure in Sarcastic Sentence Translation

The evaluation confirmed that **Sarcasm and Irony** remain the "Achilles' heel" of current NMT systems. Models consistently translate sarcastic intent as a sincere statement, losing the speaker's original emotional nuance.

11 Future Scope

To bridge the gap between the current aggregate BLEU score of 0.0448 and human-level fluency, two primary avenues for future research are proposed.

11.1 Integration of Large Language Models (LLMs)

While the NMT models evaluated here are "encoder-decoder" architectures (Transformers), moving toward **Large Language Models (LLMs)** like GPT-4 or Llama-3 offers a new paradigm.

- **Few-Shot Prompting:** LLMs can be "primed" with a few examples of sarcastic or idiomatic Konkani translations to improve their contextual understanding.
- **Zero-Shot Capabilities:** LLMs exhibit a better understanding of world knowledge, which could help in resolving the "Literalism" failure identified in Section 10.2.

11.2 Crowdsourcing and Human-in-the-loop Fine-tuning

The most sustainable way to improve low-resource translation is the expansion of high-quality data through the **Bhashini Bhasha-Daan** initiative.

- **Human-in-the-loop (HITL):** Implementing a system where native Konkani speakers verify and correct machine outputs. These "Gold Standard" corrections can then be used to fine-tune the models in real-time.
- **Dialectal Diversity:** Future datasets must intentionally include variations from across Goa (Antruz, Sashti, Bardez) to ensure the translation tools are inclusive of all Konkani speakers.

12 Conclusion

This study provided a rigorous comparative evaluation of Neural Machine Translation (NMT) systems for the Hindi-to-Konkani language pair. By benchmarking global engines against specialized regional models across a 75-sentence corpus spanning fifteen linguistic domains, we have mapped the current technical frontier for this low-resource pair.

The experimental results identify the **Anuvaad-Bhashini (IIT Bombay v1.0)** model as the top performer, achieving a leading aggregate BLEU score of **0.0448**. This model demonstrated superior structural and technical accuracy, likely due to its fine-tuning on localized Indian administrative and academic datasets. However, the consistent failure of all evaluated models—including Google, Microsoft, and IndicTrans2—to accurately translate sarcasm, irony, and deep-seated cultural idioms highlights a critical "semantic gap" in current NMT architectures.

In conclusion, while NMT has significantly improved the accessibility of Konkani, the high degree of morphological inflection and the scarcity of digital parallel corpora remain substantial hurdles. Moving forward, the results suggest that purely neural approaches must be augmented with contextual awareness and human-in-the-loop validation. Achieving human-level fluency for Konkani will require a transition from literal translation to intent-based models,

ensuring that the unique cultural and linguistic nuances of the language are preserved in the digital age.

References

- [1] S. Chandra Sekharan et al. 2023. English to Konkani Translator Using Hindi as a Pivot Language. *ResearchGate* (2023). https://www.researchgate.net/publication/371444143_English_to_Konkani_Translator_Using_Hindi_as_a_Pivot_Language
- [2] Jay Gala et al. 2023. IndicTrans2: Towards High-Quality and Accessible Machine Translation Models for all 22 Scheduled Indian Languages. *AI4Bharat, Indian Institute of Technology Madras* (2023).
- [3] IIT Bombay (CFILT). 2023. *Anuvaad-Bhashini Project v1.0: Neural Machine Translation for Indian Languages*. Technical Report. Center for Indian Language Technology.
- [4] Anoop Kunchukuttan et al. 2020. AI4Bharat-IndicNLP Corpus: Monolingual Corpora and Word Embeddings for Indic Languages. *arXiv preprint arXiv:2005.00085* (2020).
- [5] National Language Translation Mission (NLTm). 2024. Bhashini: Bridging the Language Barrier using AI/ML. <https://bhashini.gov.in/>. Ministry of Electronics and Information Technology (MeitY), Government of India.
- [6] Kishore Papineni, Salim Roukos, Todd Ward, and Wei-Jing Zhu. 2002. BLEU: a Method for Automatic Evaluation of Machine Translation. In *Proceedings of the 40th Annual Meeting of the Association for Computational Linguistics (ACL)*. 311–318.
- [7] Ashish Vaswani, Noam Shazeer, Niki Parmar, Jakob Uszkoreit, Llion Jones, Aidan N. Gomez, Łukasz Kaiser, and Illia Polosukhin. 2017. Attention Is All You Need. In *Advances in Neural Information Processing Systems (NeurIPS)*. 5998–6008.